Exact Local Exchange Profiles and Reconfiguration Barriers for Isosceles-Triangle-Free Grid Sets

Pierre-Baptiste Borges
Independent researcher
pierrebaptiste.borges@gmail.com

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Abstract

Let S be a fixed set of points in an integer grid containing no isosceles triangle, with collinear triples included. We introduce two local invariants. The individual-unlock profile $u_S(r)$ is the largest number of outside points that become individually admissible after deleting r points of S. The exchange profile $e_S(r)$ is the largest number of mutually compatible outside points that can be inserted after those deletions. We prove an exact slice identity for e_S and a general lower bound on token-addition/removal (TAR) reconfiguration barriers derived from either profile. We then describe an exact algorithm based on link-graph vertex covers, pair-forced deletion masks, and antichains of minimal masks.

For the public 112-point configuration in the 64×64 grid, our current computation gives $e_S(7) = 5 < u_S(7) = 6$, showing that individual admissibility can strictly overestimate repair capacity. For the public 164-point 100×100 configuration, an independently generated CNF and a verified DRAT proof establish $u_S(6) = 4$. At radius seven, a directly checked witness gives $e_S(7) \geq 5$, while an independently regenerated core-SAT instance and a second verified DRAT proof establish $e_S(7) < 6$. At radius eight, a directly checked $8 \to 6$ exchange and an independently audited exhaustive search give $e_S(8) = 6$; the negative search remains program-verified rather than proof-trace certified. It follows that every distinct isosceles-free set of size at least 164 deletes at least nine points of the 164-point configuration, every equal-size alternative has symmetric-difference distance at least 18, every improvement has distance at least 19, and every TAR path to an equal-or-larger distinct set descends to size at most 161. We make no claim that either public configuration is globally optimal.

Computational evidence. The radius-six and radius-seven upper bounds for the 100×100 instance have machine-verified DRAT certificates and independently regenerated reductions. The new radius-eight upper bound is an independently audited exhaustive program computation, but has no DRAT/LRAT proof trace. The results therefore have different evidence levels; Section 6 states those levels and the remaining trust boundaries.

1 Introduction

Write

$$G_n = \{0, 1, \ldots, n-1\}^2.$$

The extremal problem asks for a largest subset of G_n containing no three distinct points whose three pairwise distances include a repeated value. Thus an isosceles triangle is forbidden even when its three vertices are collinear. The problem has recently served as a benchmark for machine-assisted construction. PatternBoost reported SAT optima for small grids, a 110-point set for

$n = 64$, and a 160-point set for $n = 100$ [CEWW24]. AlphaEvolve subsequently published the 112- and 164-point configurations studied here and expressly did not claim that the latter is optimal [GGSTW25, Goo]. Geometry-aware Monte Carlo tree search has also treated this problem as one of several combinatorial-geometry benchmarks [ZZWK26].

The global and asymptotic questions remain difficult. Croot, Mao, Pohoata, Sheffer, and Yip proved an upper bound of the form

$$n^2 \exp\left(-c \frac{\log n}{\log \log n}\right)$$

for the maximum cardinality, for an absolute constant $c > 0$ [CMP$^+$26]. Counting all isosceles triangles in a regular grid is a related but distinct problem [Wu16]. Our focus is neither global optimality nor asymptotics. We ask instead how a particular large construction can be changed locally.

Local optimality and bounded swaps are standard concepts in combinatorial optimization [KM25, GHKM26]. TAR reconfiguration, buffer size, cardinality range, and minimum-cardinality paths are also established notions [dBJM16, BMP21, HOST26]. Our candidate contribution is a concrete local profile around a fixed hypergraph independent set, its translation into a TAR-barrier lower bound, and exact computations for the two public grid configurations. Link graphs, which underlie our reduction, are standard hypergraph objects; see, for example, [HT16].

The contributions of this article are as follows.

1. We define the individual-unlock profile u_S and the compatible exchange profile e_S, and identify e_S with the exact maximum on a fixed deletion slice.
2. We derive a transition-sensitive profile-to-TAR lemma. The resulting expression is a lower bound on the true reconfiguration barrier, not an assertion that the barrier itself is exact.
3. We give an exact antichain recurrence for deciding whether k outside points can coexist after at most r deletions in any 3-uniform hypergraph.
4. We report the current profiles for the AlphaEvolve $n = 64$ and $n = 100$ constructions, including the strict separation $e_S(7) < u_S(7)$ at $n = 64$ and the exact value $e_{S_{100}}(8) = 6$.
5. We archive verified DRAT proofs for the independently audited individual-unlock cell $u_{S_{100}}(6) = 4$ and for the decisive compatible upper bound $e_{S_{100}}(7) < 6$. We separately document and audit the exhaustive radius-eight computation, which does not yet have the same proof-trace evidence level.

To our knowledge after a targeted literature search, these exact profiles and the particular profile-derived expression in Lemma 3 have not been used for this grid problem. This is a provisional priority statement, not a claim that the constituent ideas—local exchange, link graphs, TAR, or SAT certification—are new.

2 The hypergraph model and local profiles

Let $H_n = (G_n, E_n)$ be the 3-uniform hypergraph in which $\{a, b, c\} \in E_n$ if and only if a, b, c are distinct and

$$\|a - b\|_2^2 = \|a - c\|_2^2, \quad \text{or} \quad \|a - b\|_2^2 = \|b - c\|_2^2, \quad \text{or} \quad \|a - c\|_2^2 = \|b - c\|_2^2.$$

All computations use these integer equalities; no floating-point geometry is needed. Let $\mathcal{F}_n$ be the hereditary family of independent sets of H_n. Most of the statements below hold for an arbitrary finite ground set V and an arbitrary hereditary family $\mathcal{F} \subseteq 2^V$.

Fix $S \in \mathcal{F}$, write $m = |S|$, and let $R \subseteq S$ be a set of deleted original vertices. Define the set of individually admissible outsiders by

$$\mathcal{U}_S(R) = \{p \in V \setminus S : (S \setminus R) \cup \{p\} \in \mathcal{F}\}.$$

Definition 1 (Individual-unlock and exchange profiles)**.** For $0 \le r \le m$, define

$$\begin{aligned} u_S(r) &= \max_{\substack{R \subseteq S \\ |R| = r}} |\mathcal{U}_S(R)|, \\ e_S(r) &= \max\{|A| : R \subseteq S,\ |R| = r,\ A \subseteq V \setminus S,\ (S \setminus R) \cup A \in \mathcal{F}\}. \end{aligned}$$

Then $e_S(r) \le u_S(r)$.

The inequality can be strict because two points can each be admissible with the retained portion of S but form a forbidden triple together with a third point, or three outsiders can themselves form a forbidden triple. The $n = 64$, $r = 7$ computation in Section 5 gives a concrete strict example.

For $X \in \mathcal{F}$, put $R_X = S \setminus X$, $A_X = X \setminus S$, and $r_X = |R_X|$. Heredity implies

$$A_X \subseteq \mathcal{U}_S(R_X),$$

since $(S \setminus R_X) \cup \{p\} \subseteq X$ for each $p \in A_X$. Consequently

$$|X| = m - r_X + |A_X| \le m - r_X + u_S(r_X). \tag{1}$$

The compatible profile gives the exact maximum rather than only an upper bound.

Proposition 2 (Exact slice identity)**.** *For every* $0 \le r \le m$*,*

$$\max_{\substack{X \in \mathcal{F} \\ |S \setminus X| = r}} |X| = m - r + e_S(r).$$

Proof. Every set in the slice decomposes uniquely as $X = (S \setminus R_X) \cup A_X$, with $|R_X| = r$ and $A_X \subseteq V \setminus S$. Maximizing $|X| = m - r + |A_X|$ over exactly these feasible pairs (R_X, A_X) is the definition of $e_S(r)$. □

We use symmetric-difference distance

$$d_\triangle(X, Y) = |X \triangle Y|.$$

In this paper, "k-swap-optimal for improvement" means that no $T \in \mathcal{F}$ with $|T| > |S|$ satisfies $d_\triangle(S, T) \le k$. This convention should be stated explicitly: other local-search papers may bound the numbers deleted and inserted separately, in which case the numerical radius must be translated rather than copied.

3 A profile-derived TAR barrier

A TAR path is a sequence $X_0, X_1, \dots, X_\ell \in \mathcal{F}$ satisfying $|X_i \triangle X_{i+1}| = 1$ for every i. For an integer $R \geq 1$, define

$$\begin{aligned} D_S^u(R) &= \max\left\{0, \max_{0\leq r<R}(r+1-u_S(r))\right\}, \\ D_S^e(R) &= \max\left\{0, \max_{0\leq r<R}(r+1-e_S(r))\right\}. \end{aligned} \tag{2}$$

Because $e_S(r) \leq u_S(r)$, one has $D_S^e(R) \geq D_S^u(R)$.

Lemma 3 (Deletion-transition TAR barrier)**.** *Let V be finite, let $\mathcal{F} \subseteq 2^V$ be hereditary, and let $S \in \mathcal{F}$ have size m. Suppose $X_0 = S, X_1, \dots, X_\ell = T$ is a TAR path and $|S \setminus T| \geq R$. Then*

$$\min_{0\leq i\leq \ell} |X_i| \leq m - D_S^u(R), \tag{3}$$

$$\min_{0\leq i\leq \ell} |X_i| \leq m - D_S^e(R). \tag{4}$$

Thus $D_S^u(R)$ and $D_S^e(R)$ are lower bounds on the loss below $|S|$ forced along every such path.

Proof. Set $r_i = |S \setminus X_i|$. We have $r_0 = 0$ and $r_\ell \geq R$. A one-vertex TAR move changes r_i by at most one. Therefore, for every $r = 0, 1, \dots, R-1$, the path contains an upward transition with $r_i = r$ and $r_{i+1} = r+1$. Such a transition removes one retained element of S, and hence $|X_{i+1}| = |X_i| - 1$.

Immediately before this deletion, inequality (1) gives

$$|X_i| \leq m - r + u_S(r) = m - (r - u_S(r)).$$

Consequently

$$|X_{i+1}| \leq m - r - 1 + u_S(r) = m - (r + 1 - u_S(r)).$$

Taking the most restrictive of these mandatory deletion transitions, together with the trivial $\min_i |X_i| \leq |X_0| = m$, yields (3). Proposition 2 replaces the pre-deletion slice bound by its exact maximum $m - r + e_S(r)$ and gives (4). □

Remark 4 (What the formula does and does not prove). Even when every value of u_S or e_S is exact, $D_S^u(R)$ or $D_S^e(R)$ is only a lower bound on the true TAR barrier. Compatibility between slices and the order of valid moves can force a deeper valley. If one has only certified upper bounds $\overline{u}(r) \geq u_S(r)$, then $\max\{0, \max_{r<R}(r+1-\overline{u}(r))\}$ is still a valid, possibly weaker, lower bound. The strict range $r < R$ in (2) is essential: a path whose target deletes at least R originals is guaranteed to make the upward transitions $r \to r+1$ for $0 \leq r < R$ without any further hypothesis. A path reaching $R+1$ makes one additional transition, so a known radius-R profile value can strengthen the maximum; we use $r < R$ deliberately because it requires only the strict prefix. The maximum with zero in (2) also covers a nonmaximal S, for which additions before the first deletion may prevent any loss below m. The configurations studied here have $u_S(0) = 0$.

Corollary 5 (Local isolation and improvement distance)**.** *Assume $e_S(0) = 0$, and let $R \geq 1$.*

(i) If $e_S(r) < r$ for $1 \leq r < R$, then every $T \in \mathcal{F} \setminus \{S\}$ with $|T| \geq |S|$ deletes at least R elements of S.

(ii) If $e_S(r) \leq r$ for $0 \leq r < R$, then every strict improvement $T \in \mathcal{F}$ with $|T| > |S|$ deletes at least R elements of S and satisfies $d_\triangle(S, T) \geq 2R + 1$.

(iii) Under the hypothesis in (i), every distinct equal-size T satisfies $d_{\triangle}(S,T) \geq 2R$.

Proof. Let $r = |S \setminus T|$ and $a = |T \setminus S|$. Feasibility gives $a \leq e_S(r)$. If $|T| \geq |S|$, then $a \geq r$, contradicting $e_S(r) < r$ when $1 \leq r < R$; the $r = 0$ case has $T = S$ because $e_S(0) = 0$. For a strict improvement, $a \geq r + 1$; hence $e_S(r) \leq r$ excludes $r < R$, and $d_{\triangle}(S,T) = r + a \geq 2R + 1$. For a distinct equal-size set, $a = r$ and the same identity gives distance at least $2R$. □

4 Exact exchange computation by antichains

This section gives the mathematical recurrence used by the radius-seven and radius-eight searches. It is independent of grid geometry and applies to a fixed independent set S in any 3-uniform hypergraph $H = (V, E)$.

4.1 Link graphs and individual covers

For $p \in V \setminus S$, define its link graph on S by

$$L_S(p) = \big(S, \{\{a,b\} \subseteq S : \{p,a,b\} \in E\}\big).$$

Then

$$p \in \mathcal{U}_S(R) \quad \Longleftrightarrow \quad R \text{ is a vertex cover of } L_S(p). \tag{5}$$

Indeed, the only possible hyperedge in $(S \setminus R) \cup \{p\}$ contains p and two retained vertices of S.

Fix a deletion radius r. Let $\mathcal{C}_r(p)$ be the family of inclusion-minimal vertex covers of $L_S(p)$ having size at most r. A branching recursion that selects an uncovered edge ab and branches on deleting a or b enumerates this family exactly. Lower bounds from edge-disjoint matchings can prune a branch without changing the answer.

4.2 Pair-forced masks and the antichain recurrence

For two outsiders $p, q \in V \setminus S$, put

$$F_S(p,q) = \{s \in S : \{p,q,s\} \in E\}.$$

Every deletion set supporting both p and q must contain the entire mask $F_S(p,q)$.

For a family $\mathcal{A} \subseteq 2^S$, write $\min_{\subseteq}(\mathcal{A})$ for its inclusion-minimal members. Define $\mathcal{M}_r(\varnothing) = \{\varnothing\}$. If $Q \subseteq V \setminus S$ has no hyperedge entirely among its elements and $p \notin Q$, define

$$\mathcal{M}_r(Q \cup \{p\}) = \min_{\subseteq}\Big\{M \cup C \cup \bigcup_{q \in Q} F_S(p,q) : M \in \mathcal{M}_r(Q),\ C \in \mathcal{C}_r(p),$$
$$\big|M \cup C \cup \bigcup_{q \in Q} F_S(p,q)\big| \leq r\Big\}. \tag{6}$$

If $Q \cup \{p\}$ contains an all-outsider hyperedge, set $\mathcal{M}_r(Q \cup \{p\}) = \varnothing$.

Proposition 6 (Exactness of the antichain recurrence)**.** *For every $Q \subseteq V \setminus S$, $\mathcal{M}_r(Q)$ is precisely the antichain of inclusion-minimal sets $M \subseteq S$ of size at most r such that $(S \setminus M) \cup Q$ is independent. Consequently,*

$$e_S(r) \geq k \quad \Longleftrightarrow \quad \textit{there exists } Q \subseteq V \setminus S, |Q| = k,\ \mathcal{M}_r(Q) \neq \varnothing.$$

Here a supporting mask of size less than r may be padded with arbitrary additional deletions from S.

Proof. The claim is immediate for $Q = \varnothing$. Suppose it holds for Q. Any forbidden triple in $(S \setminus M') \cup (Q \cup \{p\})$ is of one of four types. A triple contained in S is impossible because S is independent. A triple containing p and two vertices of S is excluded exactly when M' contains a cover in $\mathcal{C}_r(p)$, by (5). A triple containing p, q and one vertex $s \in S$ is excluded exactly when $F_S(p, q) \subseteq M'$. Conditions involving old elements of Q are already encoded by a member $M \in \mathcal{M}_r(Q)$. Finally, a triple entirely in $Q \cup \{p\}$ cannot be repaired by deleting elements of S and is rejected explicitly.

Thus the unions in (6) are exactly the supporting masks of size at most r, before removal of supersets. Taking inclusion-minimal members preserves existence and yields precisely the claimed antichain. The final equivalence follows from the definition of $e_S(r)$ and the fact that independence is preserved under additional deletions. □

The implementation first constructs a graph on eligible outsiders, retaining an edge only when its two endpoints have a common supporting mask of size at most r. A feasible k-set must be a k-clique in this graph, so a clique-style DFS provides an efficient necessary prefilter. Pairwise compatibility is not sufficient: at every node the DFS carries the full antichain from (6) and checks all-outsider triples.

5 Computational results

The two main configurations are the public AlphaEvolve outputs [GGSTW25, Goo]. We denote the 164-point set in G_{100} by S_{100} and the 112-point set in G_{64} by S_{64}. Section 6 separates theorem-level deductions from the current evidence for each computed cell.

5.1 The 164-point construction in G_{100}

The current profile is

r	0	1	2	3	4	5	6	7	8
$u_{S_{100}}(r)$	0	0	1	1	2	3	4	≥ 5	≥ 6
$e_{S_{100}}(r)$	0	0	1	1	2	3	4	5	6

Only the displayed lower bounds are claimed for $u_{S_{100}}(7)$ and $u_{S_{100}}(8)$; their exact values remain open. Time-limited decisions at both radii returned UNKNOWN, which is not an upper bound. There is a directly checked $7 \to 5$ exchange:

$$
\begin{aligned}
R &= \{(92,51),(96,97),(7,51),(8,93),(26,40),(73,40),(77,95)\},\\
A &= \{(95,98),(89,56),(75,81),(10,56),(77,99)\}.
\end{aligned}
$$

Exact integer geometry verifies that $(S_{100} \setminus R) \cup A$ is independent. The antichain DFS also reports $e_{S_{100}}(7) < 6$, with 15,227 search states and 47.762 seconds in the logged run. The decisive upper bound no longer rests on that traversal alone: a separately encoded no-symmetry core CNF and its DRAT proof have passed independent regeneration and proof checking, as detailed in Section 6.

At radius eight, an independently checked exchange is

$$
\begin{aligned}
R = \{&(26,59),(7,48),(73,59),(22,4),\\
&(92,48),(77,4),(96,2),(3,2)\},\\
A = \{&(4,1),(10,43),(22,0),(77,0),(89,43),(95,1)\}.
\end{aligned}
$$

Exact integer geometry verifies that the resulting 162-point set is independent. An exhaustive antichain search rejects seven additions even in a necessary-condition relaxation that omits all constraints on triples of outsiders and disables color pruning. Section 6 records the independent input regeneration, search counts, and remaining program-verification boundary.

Computational claim 7 (Audited local consequences for S_{100})**.** *The radius-eight computation described in Section 6 implies:*

(i) every $T \neq S_{100}$ with $|T| \geq 164$ deletes at least nine points of S_{100};

(ii) every distinct 164-point set has $d_\triangle(S_{100}, T) \geq 18$;

(iii) every set of size at least 165 has $d_\triangle(S_{100}, T) \geq 19$;

(iv) S_{100} is 18-swap-optimal for improvement under the symmetric-difference convention of Section 2; and

(v) every TAR path from S_{100} to a distinct set of size at least 164 contains a set of size at most 161, so no such endpoint lies in the same component when TAR is restricted to sets of size at least 162;

(vi) the component of S_{100} in the TAR graph induced by sets of size at least 163 is exactly the star with center S_{100} and leaves $S_{100} \setminus \{s\}$ for $s \in S_{100}$.

Proof. For $1 \leq r \leq 8$, the displayed exchange values satisfy $e_{S_{100}}(r) < r$. Corollary 5 with $R = 9$ gives (i)–(iv). For (v), the endpoint must delete at least nine original points by (i), and Lemma 3 applies. In fact the already computed slice $r = 3$ has $e_{S_{100}}(3) = u_{S_{100}}(3) = 1$. Immediately after the mandatory transition from three to four deleted originals, the path therefore has size at most $164 - 3 + 1 - 1 = 161$. Equivalently, the known profile entries give $D^u_{S_{100}}(9) = D^e_{S_{100}}(9) = 3$: the open values satisfy $u_{S_{100}}(7) \geq 5$ and $u_{S_{100}}(8) \geq 6$, so their contributions to D^u cannot exceed 3.

For (vi), $e_{S_{100}}(0) = 0$ means that no outsider can be added directly to S_{100}, so its only TAR neighbors are its 164 single deletions. Such a leaf has size 163: a further deletion violates the threshold, adding an outsider is impossible because $e_{S_{100}}(1) = 0$, and restoring its unique missing record point returns to S_{100}. $\square$

Exploratory radius-nine frontier. The next improvement decision is $e_{S_{100}}(9) \geq 10$. A bounded exact DFS returned UNKNOWN after 580.099 seconds, 233,472 nodes, and 91,148,661,627 cover tests; three deterministic heuristics evaluated 8,446,705 deletion masks without finding a $9 \to 10$ witness. These are timeouts and sampled failures, not an upper bound. They did produce a directly verified lower bound $u_{S_{100}}(9) \geq 7$. For one archived mask the seven unlocked outsiders are pairwise compatible, but

$$(4, 98),\ (75, 81),\ (77, 99)$$

form an isosceles triple: the first point has squared distance 5330 to each of the other two. Hence the maximum compatible subset of this particular seven-point list is six. This gives a concrete radius-nine illustration that pair compatibility alone does not capture all repair obstructions; globally, both profile values at radius nine remain open beyond their lower bounds.

None of these statements asserts that 164 is the maximum cardinality in G_{100}. They describe the neighborhood of this one public construction.

5.2 The 112-point construction in G_{64}

For S_{64}, the current exact computation is

r	0	1	2	3	4	5	6	7
$u_{S_{64}}(r)$	0	1	2	2	4	4	5	6
$e_{S_{64}}(r)$	0	1	2	2	4	4	5	5

In particular,

$$e_{S_{64}}(7) = 5 < 6 = u_{S_{64}}(7).$$

This strict separation demonstrates why the individual-unlock profile cannot in general be treated as the capacity of a simultaneous repair. The radius six compatible optimum is also five: $e_{S_{64}}(6) = u_{S_{64}}(6) = 5$.

The configuration is not isolated among equal-size sets. The exchange

$$S'_{64} = \big(S_{64} \setminus \{(56,2)\}\big) \cup \{(62,2)\}$$

is valid, so $d_{\triangle}(S_{64}, S'_{64}) = 2$. On the other hand, any strict improvement must delete at least eight original points and therefore has symmetric-difference distance at least 17. Thus S_{64} is 16-swap-optimal for improvement under our convention. Moreover,

$$D^{u}_{S_{64}}(8) = 2, \qquad D^{e}_{S_{64}}(8) = 3.$$

Every TAR path from S_{64} to an improvement consequently contains a set of size at most 109. The compatible profile gives a strictly stronger barrier than the individual profile on this instance.

6 Certification and evidence levels

Computer-assisted extremal results require two separate arguments: the CNF or search instance must faithfully encode the mathematical problem, and the claimed solver outcome must be checkable. A second solver using the same generator is useful testing but is not an independent certificate. Certified SAT workflows for combinatorial mathematics motivate our standard here [HKM16, CFHH+17, Sze26].

6.1 Independently audited cell $u_{S_{100}}(6) = 4$

The lower bound has an explicit witness

$$R = \{(26,59), (7,48), (73,59), (92,48), (96,2), (3,2)\},$$
$$\mathcal{U}_{S_{100}}(R) = \{(4,1), (10,43), (89,43), (95,1)\}.$$

An independent integer-geometry implementation verifies that these four and only these four points are unlocked by R.

For the upper bound, a separate generator extracted the record from the public notebook, checked its validity, rebuilt all conflicts by distance rings, and constructed the decision instance "six deletions unlock at least five outside points." The grid has 9,836 outsiders and 154,440 link-graph edges in total. A simple certificate of seven pairwise disjoint conflict edges excludes 9,099 outsiders, because six deleted record vertices cannot cover such a matching. The independently checked matching file leaves 737 conservative candidates in the CNF.

The archived instance has 5,703 variables and 24,775 clauses. Its identity is fixed by the following artifact paths and hashes:

CNF `research/audit/r6_t5_matching.cnf`
SHA-256: `f72229bacc773b19bf78209134cc666496854e60d1a515b4809ae2deeca23651`

DRAT proof `research/audit/r6_t5_matching.drat`
SHA-256: `149d3592fe5a2d6afccc05905b45ddc638947d537df6bc99865b9d190b20be9a`

The CNF is 381,497 bytes. CaDiCaL 1.9.5 at commit `146207318796f094dcded87349a64f0c6927309e` produced a 291,778,586-byte binary DRAT proof. `drat-trim` at commit `2e3b2dc0ecf938addbd779d42877b6ed69d9a985` returned `s VERIFIED` in 228.846 seconds. The proof manifest records all commands, hashes, versions, and checker output.

This proof validates the archived CNF. Confidence in the mathematical translation comes from the independent generator, direct witness checks, matching certificates, exhaustive small cardinality tests, and a written proof of the encoding. It is not a formal proof of the geometry-to-CNF generator. A conversion of the full instance to LRAT followed by a CakeML-produced checker has not been completed because of local storage constraints; only the checker chain on a tiny smoke instance has passed. The manuscript therefore does not claim end-to-end formal verification.

6.2 Machine-verified radius-seven exchange bound

For S_{100} at $r = 7$, there are 1,324 eligible outsiders. Of the $\binom{1324}{2} = 875{,}826$ pairs, pair preprocessing rejects 863,884 and retains 11,942. An independently written pair-cache regenerator agrees with those decisions and hashes. The antichain DFS reports UNSAT for six additions after 15,227 nodes, 1,013,058 family extensions, and 475,262 generated masks. This is now corroborative rather than the sole upper-bound evidence.

Any six-vertex clique lies in the 5-core of the pair-compatibility graph. The exact core has 392 candidates and 11,054 compatible edges. A second SAT encoding imposes the one-outsider cover clauses, every two-outsider forced deletion mask, and all three-outsider incompatibilities. Independent standard-library programs regenerate the eligible candidates, pair graph, 5-core, and all 200,796 ternary no-goods. The resulting no-symmetry CNF has 7,386 variables and 308,590 clauses:

Core CNF `research/audit/radius7_core_r7_k6_nosym.cnf`
SHA-256: `2461ee72b797a636de8154422544c2c3673e43d9c83e069f41a1c434eef26bba`

DRAT proof `research/audit/radius7_core_r7_k6_nosym.drat`
SHA-256: `6a845c1af080a63661599f9633910b170293502c9319688cf55829ca1ba37c3b`

The CNF is 4,855,298 bytes. CaDiCaL returned UNSAT with exit code 20 and wrote a 25,282,853-byte proof. Independent `drat-trim` verification returned `s VERIFIED` with exit code 0 in 26.669 seconds; its core contained 252,110 of the 308,590 input clauses and used 22,191,309 resolution steps, with zero RAT lemmas in the core. Together with the directly checked five-addition witness, this establishes $e_{S_{100}}(7) = 5$ at the machine-verified evidence level.

The DRAT checker proves that the archived CNF is unsatisfiable. The independent regenerators and the human proof of the reduction connect that CNF to the geometric exchange problem, but this connection is not formally verified in a proof assistant. The full proof has not been converted to LRAT, and neither the manuscript nor the artifacts have been peer reviewed. "Machine verified" is therefore the accurate description; "formally verified end-to-end" is not.

6.3 Independently audited radius-eight exhaustive search

For S_{100} at $r = 8$, the search input has 2,036 eligible outsiders and 150,848 inclusion-minimal one-outsider covers. Its pair preprocessing stores 331,300 nonempty forced-deletion masks and retains 35,950 of the 2,071,630 outsider pairs. A standard-library audit, using an alternative vertex-cover enumerator and direct squared-distance geometry, independently regenerated the record, every candidate and cover, every forced mask, and every pair decision. All counts and decisions matched the binary input and JSON cache exactly.

The positive side is the $8 \to 6$ witness displayed above, checked by a separate exact-integer cubic enumeration of all triples in the resulting 162-point set. For the upper bound, the C++ implementation of Proposition 6 exhaustively rejected a target of seven. A deliberately weaker replay removed every all-outsider-triple constraint and disabled the coloring bound. It still returned UNSAT, after 78,031 nodes, 10,009,190 antichain extensions, 5,388,204,289 cover-union tests, and 3,217,876 generated masks. Every genuine seven-point exchange is feasible in this relaxation, so this negative result yields $e_{S_{100}}(8) < 7$ without relying on either the all-outsider-triple code or color pruning. Together with the witness, it gives $e_{S_{100}}(8) = 6$.

The source was independently recompiled and the decisive relaxed run replayed with identical structural counters. Regression tests reproduce the certified radius-seven outcome; sanitizer and randomized differential tests exercise the mask recurrence and coloring bound. These checks materially narrow the trust boundary, but they are not a portable proof trace. No DRAT/LRAT certificate or proof-assistant verification exists for the radius-eight upper bound. We therefore describe it as an exact exhaustive computation, independently audited and cross-checked, rather than as formally or proof-trace verified.

The S_{64} runs and the remaining table cells have executable witnesses and current exact-solver logs, but do not yet have a uniform suite of independently checked proof certificates comparable to the decisive S_{100} radius-six and radius-seven artifacts. They are accordingly reported as reproducible computations rather than proof-trace-certified results.

7 Algorithmic interpretation

The profiles have an immediate use in destroy-and-repair search. After deleting r elements of an incumbent S, no feasible repair can insert more than $e_S(r)$ outsiders, and the cheaper profile $u_S(r)$ is a safe upper bound. A radius satisfying $u_S(r) \le r$ cannot directly improve S. Consequently, a solver can skip such radii when it seeks a one-shot improvement, or deliberately treat them only as transit states in a non-monotone search.

For S_{100}, the audited values exclude a direct improvement through radius eight. Any successful one-shot destroy-and-repair search must therefore destroy at least nine incumbent points; the next record-improvement decision is whether ten outsiders can be inserted after nine deletions. In a global model seeking at least 165 points, let x_s indicate whether the record point s is retained. The same result gives the valid incumbent-overlap cut

$$\sum_{s \in S_{100}} x_s \le 155.$$

It excludes all $\sum_{r=0}^{8} \binom{164}{r} = 11{,}494{,}182{,}836{,}236$ deletion masks in the first nine shells before any repair variables are considered. For S_{64}, the difference between u and e at radius seven shows that ranking a destroy set only by the number of individually unlocked points can be misleading: the best unlocked collection need not contain a compatible repair of the same size.

These observations are algorithmic consequences, not evidence that the profiles predict heuristic performance in general. Establishing such a link would require multiple nonisomorphic constructions, repeated solver runs, and an experimental protocol fixed before evaluating the outcomes.

8 Reproducibility and provenance

The accompanying research directory is intended to be the reproducibility package. The key files are:

- `research/audit/AUDIT.md`, the independent radius-six audit;
- `research/audit/proof_manifest.json` and `research/audit/certificate_checksums.sha256`, recording the instance, proof, tools, and hashes;
- `research/audit/verify_certificate.sh`, the one-command replay of checksums, matching witnesses, and DRAT verification;
- `research/radius7_exchange_dfs.py` and `research/radius7_pair_compatibility.py`, the exchange solver and pair preprocessor;
- `research/radius7_dfs_n100_r7_k5.txt` and `research/radius7_dfs_n100_r7_k6.txt`, the radius-seven witness and current upper-bound log;
- `research/radius7_dfs_n64_r6_k5.txt`, `research/radius7_dfs_n64_r7_k5.txt`, and `research/radius7_dfs_n64_r7_k6.txt`, the compatible $n = 64$ logs;
- `research/audit/radius7_pair_cache_audit.json`, the independent pair-cache audit;
- `research/audit/radius7_core_audit.json`, the independent core and ternary-no-good regeneration;
- `research/audit/radius7_core_r7_k6_nosym.cnf` and `research/audit/radius7_core_r7_k6_nosym.drat`, the certified radius-seven upper-bound instance and proof;
- `research/radius8_cpp_antichain_dfs.cpp` and `research/radius8_cpp_REPORT.md`, the radius-eight solver and detailed computational report;
- `research/radius8_cpp_n100_r8_k6.json`, `research/radius8_cpp_n100_r8_k6_verify.json`, and `research/radius8_cpp_n100_r8_k7_relaxed_nocolor.json`, the positive witness, direct check, and decisive relaxed upper-bound run;
- `research/audit/radius8_cpp_inputs_independent.py`, `research/radius8_cpp_input_audit.json`, and `research/audit/RADIUS8_CPP_AUDIT.md`, the independent radius-eight input and implementation audits;
- `research/audit/verify_radius8_computation.sh`, the one-command checksum, witness, differential-test, rebuild, and decisive replay wrapper;
- `research/RESULTS_MANIFEST.json` and `research/audit/verify_frontier_bundle.sh`, the machine-readable claim/evidence snapshot and its fast cross-check wrapper;
- `research/RADIUS9_THEORY_NOTE.md`, the transition-barrier, incumbent-cut, and local TAR-component deductions;

- `research/audit/verify_tar_transition_theorem.py`, an exhaustive finite-model regression over all hereditary families on ground sets of size at most four (a check of edge cases, not a substitute for the proof);
- `research/radius9_FRONTIER.md`, `research/radius9_cpp_n100_r9_k10.json`, and `research/radius9_alt_search_20260827.json`, the explicitly UNKNOWN radius-nine frontier and bounded-search diagnostics;
- `research/radius9_u7_witness.json`, `research/radius9_u7_witness_verify.json`, and `research/audit/radius9_mask_direct_check.py`, the radius-nine individual-unlock witness and direct exact-integer checks.

From the repository root, the radius-six certificate and its matching witnesses are replayed by

```
research/audit/verify_certificate.sh
```

The exact radius-seven DFS can be rerun, subject to the documented Python environment, by

```
python research/radius7_exchange_dfs.py \
  --n 100 --radius 7 --target 6 \
  --pair-cache research/radius7_paircompat_n100_r7.json \
  --seconds 300 \
  --output research/radius7_dfs_n100_r7_k6.replay.txt
```

The independently checkable proof is replayed by

```
research/audit/tools/drat-trim/drat-trim \
  research/audit/radius7_core_r7_k6_nosym.cnf \
  research/audit/radius7_core_r7_k6_nosym.drat
```

The required success line is `s VERIFIED`. Replaying the DFS alone is not a certificate; replaying the DRAT checker verifies the archived CNF, while the reduction audit remains a separate part of the trust chain.

The radius-eight checks and decisive replay are run together by

```
research/audit/verify_radius8_computation.sh
```

Passing `-full-input-audit` additionally regenerates every candidate, cover, forced mask, and pair decision. Equivalently, the decisive relaxation alone can be rebuilt and replayed by

```
g++ -O3 -DNDEBUG -std=c++20 \
  -DR8CPP_RELAX_OUTSIDE_TRIPLES -DR8CPP_NO_COLOR_PRUNING \
  research/radius8_cpp_antichain_dfs.cpp -o /tmp/r8-relaxed
/tmp/r8-relaxed \
  research/radius7_pair_input_n100_r8.bin \
  research/radius7_paircompat_n100_r8.json \
  7 600 /tmp/r8-k7-relaxed.json
```

The expected status is UNSAT with exit code 20. This is a deterministic replay of the audited exhaustive program, not an independently checkable UNSAT certificate.

The AlphaEvolve source was pinned at repository commit `8f447457957deac61e28bf1676746f0753b3b2f8`. The source notebook has SHA-256 `653accd24369672f1cc757b203dae311f1489895a1cb8e6242d3308fe657891d`, and the canonical serialized S_{100} coordinates have SHA-256 ffe9583db47dbcc7e29e12a7d148fb1388827ac9917d1aefa7eb5fee89b4e7d6. The accompanying archive preserves the sources, environment specification, witnesses, CNFs, proofs, checkers, manifests, and manuscript under immutable hashes.

The source repository and frozen artifact release are available at https://github.com/Liftof/local-exchange-profiles and https://github.com/Liftof/local-exchange-profiles/releases/tag/v1.0.0.

9 Limitations and open problems

The present study has several material limitations.

1. The exact values $u_{S_{100}}(7)$, $u_{S_{100}}(8)$, and both profiles at radius nine are open. A timeout or a failed heuristic search is neither evidence of satisfiability nor a certified upper bound.

2. The upper bound $e_{S_{100}}(7) < 6$ has a verified DRAT proof, but the geometry-to-CNF reduction is audited software rather than an end-to-end theorem in a proof assistant. No full LRAT/CakeML check has been performed.

3. The upper bound $e_{S_{100}}(8) < 7$ is an independently audited and replayed exhaustive computation, but it has no DRAT/LRAT proof trace. It therefore retains the C++ implementation, compiler, and hardware as part of its trust base.

4. Certification is uneven across the two tables. The decisive S_{100} radius-six and radius-seven bounds have verified DRAT chains; other upper-bound cells need similarly portable evidence.

5. The formulas D^u_S and D^e_S lower-bound a true TAR barrier but do not generally determine it. Computing exact max–min reconfiguration values is a separate problem.

6. Two principal configurations are insufficient for a general empirical law. Profiles for all SAT-optimal small grids and for multiple nonisomorphic large configurations would test whether rigidity correlates with size or search difficulty.

7. No result here proves $C(64) = 112$, $C(100) = 164$, or any other global optimum. The asymptotic bound in [CMP+26] does not close these finite cases.

8. The novelty statement is based on a targeted search of primary sources. It should be checked against MathSciNet and zbMATH and discussed with domain experts before asserting priority.

Promising next questions include producing a portable proof trace for the radius-eight rejection, converting the existing proofs to LRAT, solving $u_{S_{100}}(7)$ and $u_{S_{100}}(8)$, profiling small globally optimal instances, comparing nonisomorphic sets of equal size, and deciding the next record frontier $e_{S_{100}}(9) \geq 10$. The radius-nine counters indicate that caching each complete deletion mask's unlocked-candidate bitset is a concrete next optimization: it can replace repeated minimal-cover scans by bitset intersections. A more theoretical direction is to characterize hypergraph classes for which the profile-derived TAR bound is attained.

Use of computational tools

Generative language-model tools, including OpenAI Codex, assisted computational exploration, code development, literature search, and drafting. The human author is responsible for the manuscript, its claims, references, and accompanying artifacts.

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The public configurations studied here originate in the AlphaEvolve repository and associated paper [GGSTW25, Goo].

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